UCT HONOURS · ARTIFICIAL LIFE · LIFE ENGINE

DIGGING THEIR OWN GRAVES



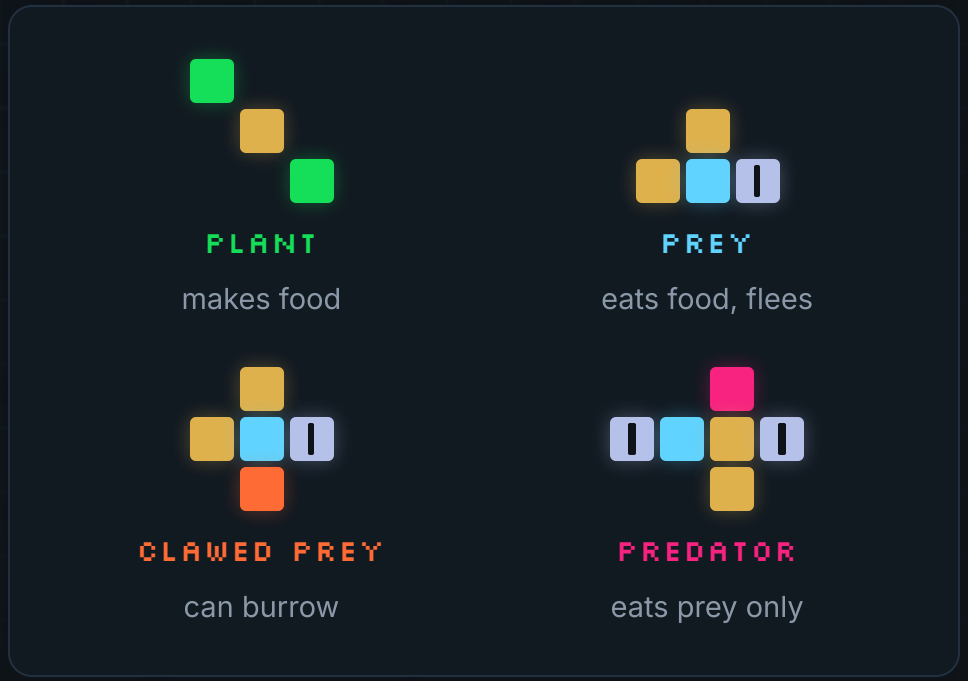
Burrowing, soil and extinction: can a trait that saves each individual doom the population?

THE ORGANISM

A world of cells, soil and one new trait

- **Organisms are built from cells.** Each cell type does one job, and offspring mutate by adding, removing or changing a cell.
- **Soil fertility.** Plants are only born where the soil under them averages at least 7 of 10. Unoccupied soil slowly regains fertility.
- **The claw cell.** A prey with a claw that sees a predator **burrows**: it is invisible for 100 ticks and doesn't age, but each burrow strips 10 fertility from the soil under it.
- **Roles come from the body.** A killer cell and no producer makes a predator; a producer that can't move is a plant; everything else is prey.

producer mouth mover eye killer claw food



Starting organisms as built in the simulator (cell layouts from RandomOrganismGenerator)

THE BIOLOGY

Evolutionary suicide: when adaptation causes extinction

Natural selection favours whatever helps individuals leave more offspring, **even when it harms the population**. If a favoured trait degrades something everyone depends on, selection can drive its own carriers extinct.

Gyllenberg & Parvinen (2001), *Bull. Math. Biol.*; Rankin & López-Sepulcre (2005), *Oikos*, "Can adaptation lead to extinction?"; Hardin (1968), "The tragedy of the commons", *Science*.

A real echo: burrowers reshape soil

Burrowing animals are ecosystem engineers. Introduced European rabbits in Australia, for example, burrow and graze so heavily that they strip vegetation and drive soil erosion.

Our hypothesis

1 · INDIVIDUAL GAIN
Burrowing hides a prey from predators

2 · SELECTION
Claws spread through the prey

3 · SHARED COST
Every burrow drains the soil plants need

4 · COLLAPSE
Plants fail, food runs out, burrowers die

The test: does the population survive **without** the trait, and only collapse **with** it?

Mis-adaptation = a trait that is favoured yet ruinous · failure to adapt = no suitable trait in time

THE EXPERIMENT

Nearly 2,000 simulations, run like a field study

- **Same code, no screen.** A headless runner drives the simulator's own code, so hundreds of runs go in parallel.
- **Seeded layouts, random lives.** Soil and starting positions come from seeds; everything after that is stochastic, so replicates differ.
- **Controls, like knockout experiments:** the same starting layouts with one thing changed.
  - claws **seeded** at the start, or left to **evolve** by mutation
  - burrowing **switched off**, or **harmless** to the soil
  - soil density, patch size, regrowth; predator strength; plant regrowth

Study	Focus	Runs
Tarique	evolved claws, soil cost, predators	456
Nik	evolved claws, soil settings, maps	170+
Jenna	plant regrowth, plant death	936
Josh	seeded vs evolved claws, soil layouts	380

Plus browser runs. 100,000–250,000 ticks each. Proportions compared with Fisher's exact test.

Settings differ slightly between studies; each comparison is made within one study

QUESTION 1 · DID AN EXTINCTION EVENT OCCUR?

Yes, at every level, and often

167/380

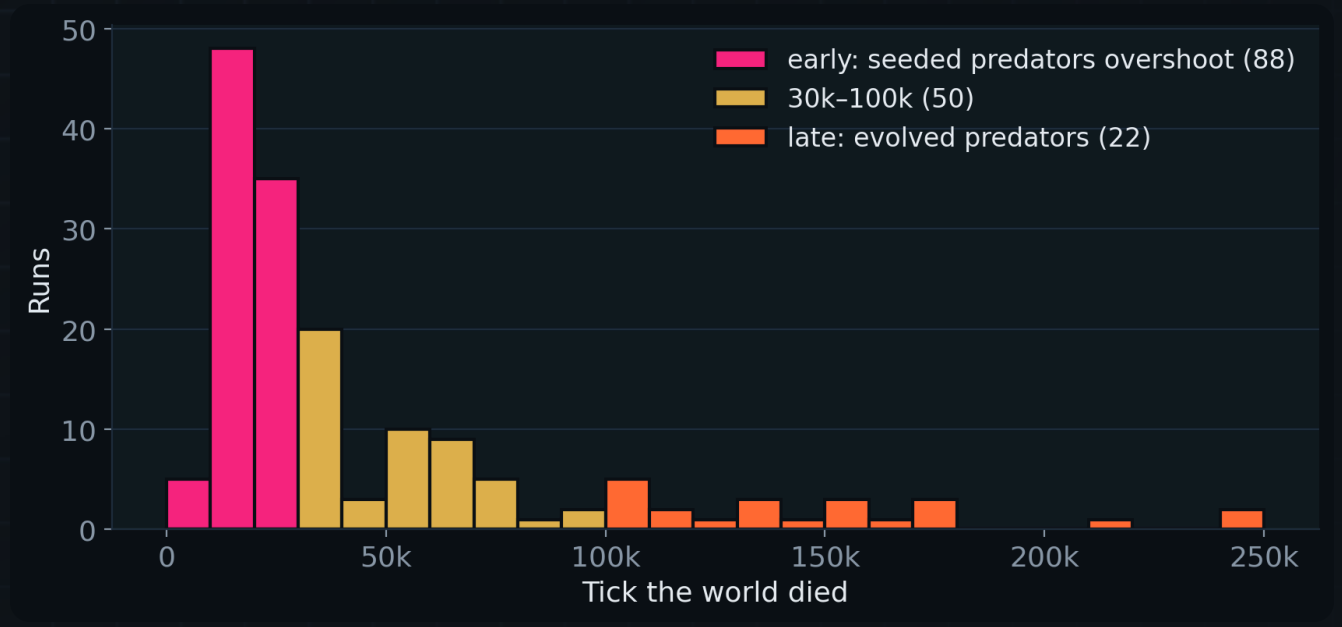
of our headless runs ended with **every organism dead**; in Nik's sweep, 111 of 111

21/40

runs where evolved claws took over and then **every prey died** (Tarique)

LINEAGES

The seeded predators died out in **every** batch run; seeded burrowing prey died out in every run with all the claw's benefits switched off



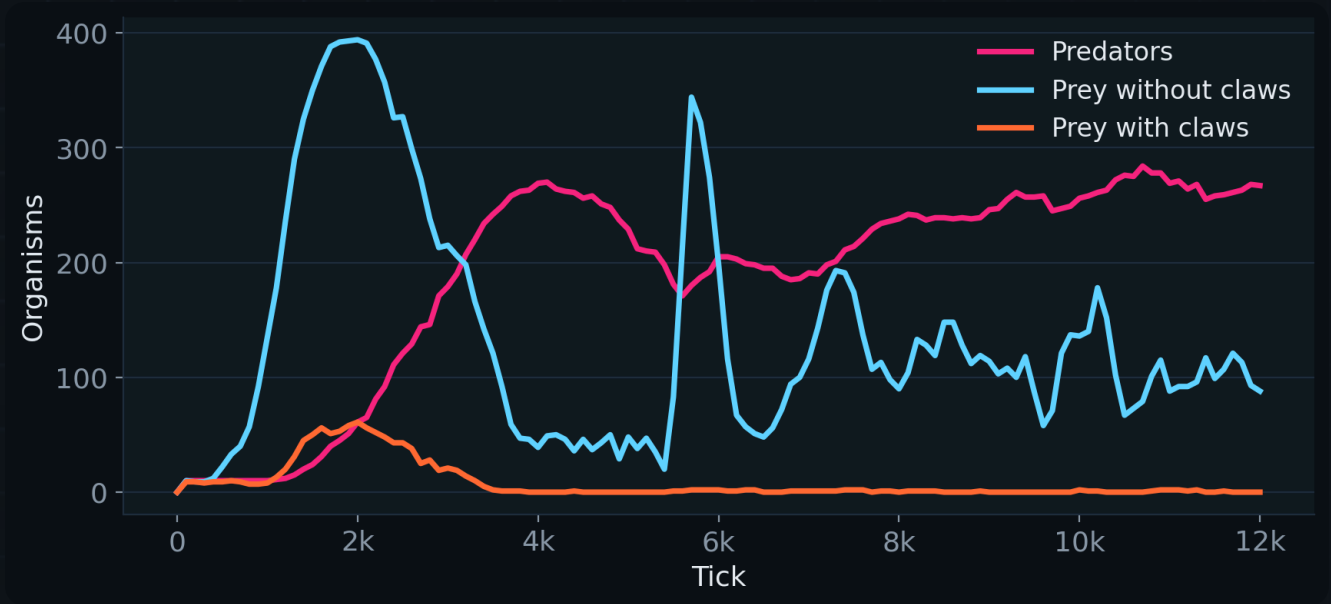
When 160 worlds died (batches with seeded claws, 300 runs). Most die early, a few after 100k ticks of apparent stability.

Whole-world extinction usually ends with plants: with no animals to eat it, food piles up and blocks plant offspring

FAILURE TO ADAPT

Most worlds die before evolution can act

- **The first predator wave.** Seeded predators feed on the first prey boom, multiply about 25×, and overshoot.
- **88 of 160** world extinctions came before 30,000 ticks. When the prey died, **82%** of predators were still the seeded ones.
- Claws appear within ~1,200 ticks, but prey were usually gone by **5,500**, while claws were on only ~4% of them (Nik).
- With all the claw's benefits off, the seeded burrowers bred at half the rate of other prey and were the first to vanish.

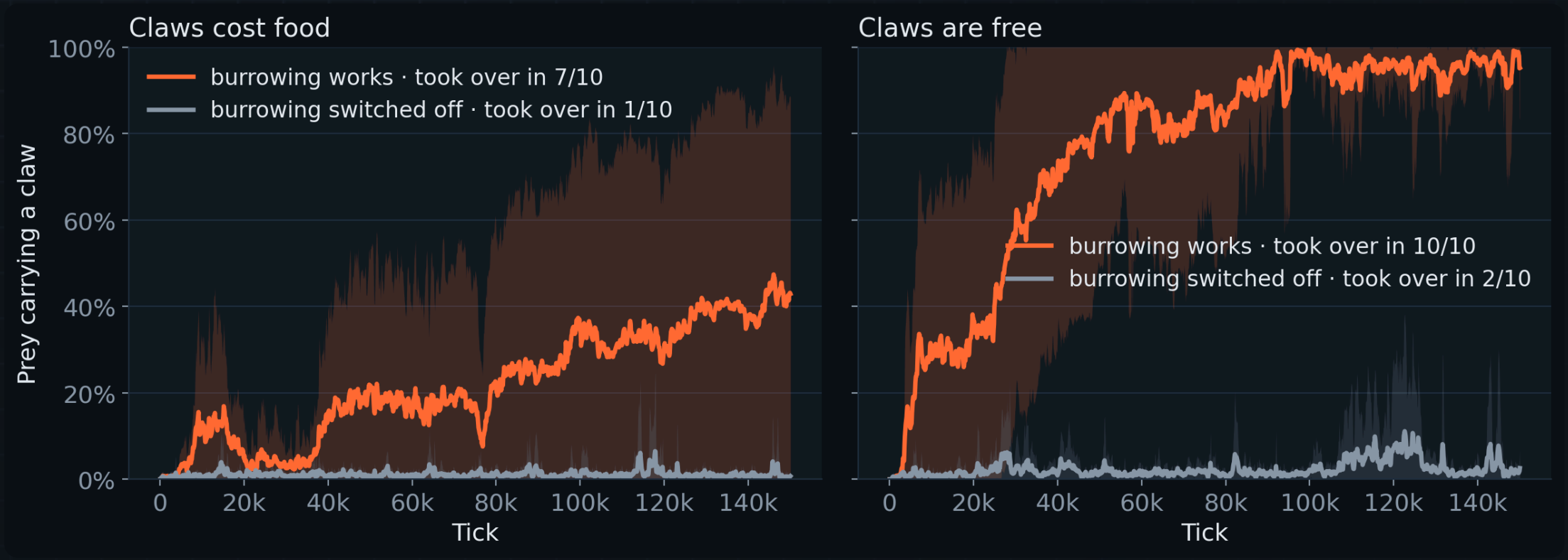


Browser run, claw benefits off: predators boom and the clawed prey are gone by tick 4,000.

Ecological overshoot, not evolution: the prey had no time to adapt

MIS-ADAPTATION

Given time, the claw evolves, because it works



No organism starts with a claw. It appears by mutation in **every run** (80 of 80), within a median of about 1,000 ticks.

It spreads only when burrowing works: 7/10 take-overs against 1/10 when claws cost food ($p = 0.02$), 10/10 against 2/10 when free ($p = 0.0007$). Selection favours the claw *for* burrowing.

Josh's claw-evolution batch · 10 runs per setup · predator lifespan multiplier 2 · bands are ±1 std

QUESTION 2 · MIS-ADAPTATION OR FAILURE TO ADAPT?

MIS-ADAPTATION …and then the burrowers dig their own grave

8K	8K → 15K	→ 20K	→ 57K	65K
Evolved claws take over the prey	Burrows drain the soil to 74%	Plants crash from 468 to 35	Clawed prey dwindle and die	Nothing is left

21/40 VS 0/20

prey extinct after claws took over, damaging vs **harmless** burrowing (Tarique, p = 0.00002). Without working claws, prey survived 17/20.

11/11 VS 0/5

burrower populations extinct after spreading, with vs without soil damage (Nik, p ≈ 0.0002).



One run (claws evolved, free, burrowing on). The soil recovers after the burrowers die, too late for them.

QUESTION 2 · MIS-ADAPTATION OR FAILURE TO ADAPT?

Both, depending on how far evolution gets

FAILURE TO ADAPT

The common route

- Strong seeded predators, prey with no defence yet
- Collapse within ~30k ticks, before the claw spreads
- Happens just as often with burrowing switched off (Nik's controls)

MIS-ADAPTATION

Evolutionary suicide, when the claw gets a chance

- The claw evolves and spreads **because** it protects (vs burrowing off)
- It kills only when burrowing damages the soil (vs harmless burrowing)
- Prey survive without the claw: the collapse is the trait's doing

A second, subtler route

Clawed prey that grow a killer cell can become predators. In 9 of 22 late collapses, the predator that wiped out the prey had evolved **from the prey's own lineage**, and in about 10 a newly evolved predator spread and the world died within a few thousand ticks. Suggestive of evolutionary suicide, but not yet proven by replay.

Not yet measured: whether clawed prey still out-reproduce clawless prey as the population declines

QUESTION 3 · ENVIRONMENT VS THE INITIAL SEEDED SPECIES?

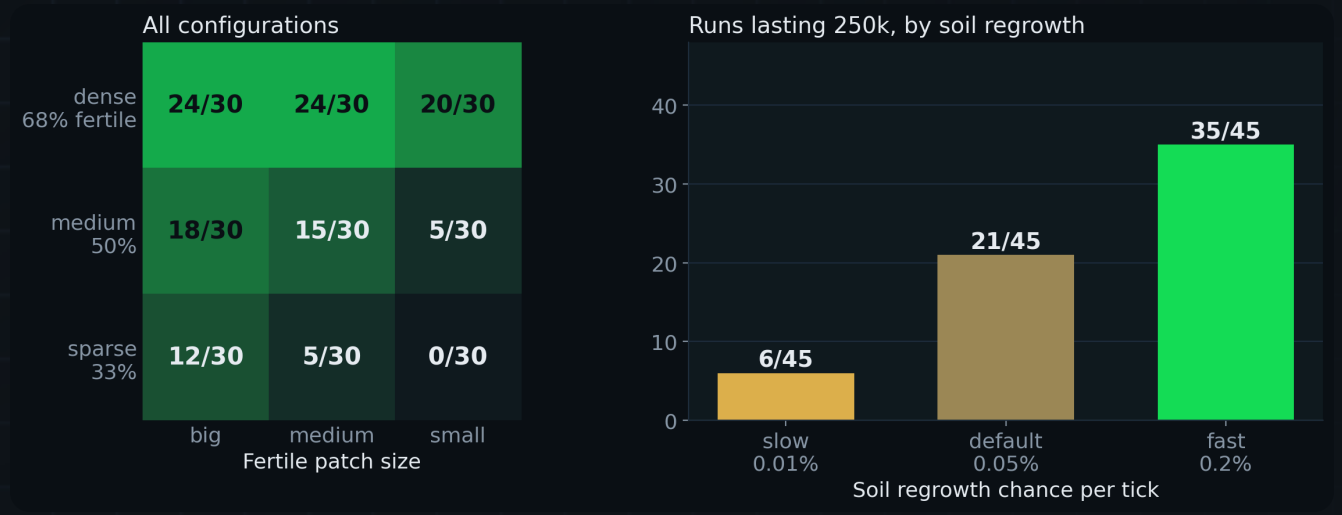
The seeded species decide whether burrowers get a chance; the environment decides what happens to them

SEEDED SPECIES

- **Predator strength** sets the first wave: at lifespan ×3, predators killed all prey in 7/8 runs even without burrowing; ×2 lets prey survive (Tarique)
- **Seeding claws** lets prey skip the wait and survive the first wave
- The seeded predators **never last**: evolved predators replace them

ENVIRONMENT

- **Soil** density, patch size and regrowth (right)
- **Plant regrowth** rescues worlds from the claw (Jenna)
- **Map luck**: the pattern appeared on one soil map, not another (Nik)

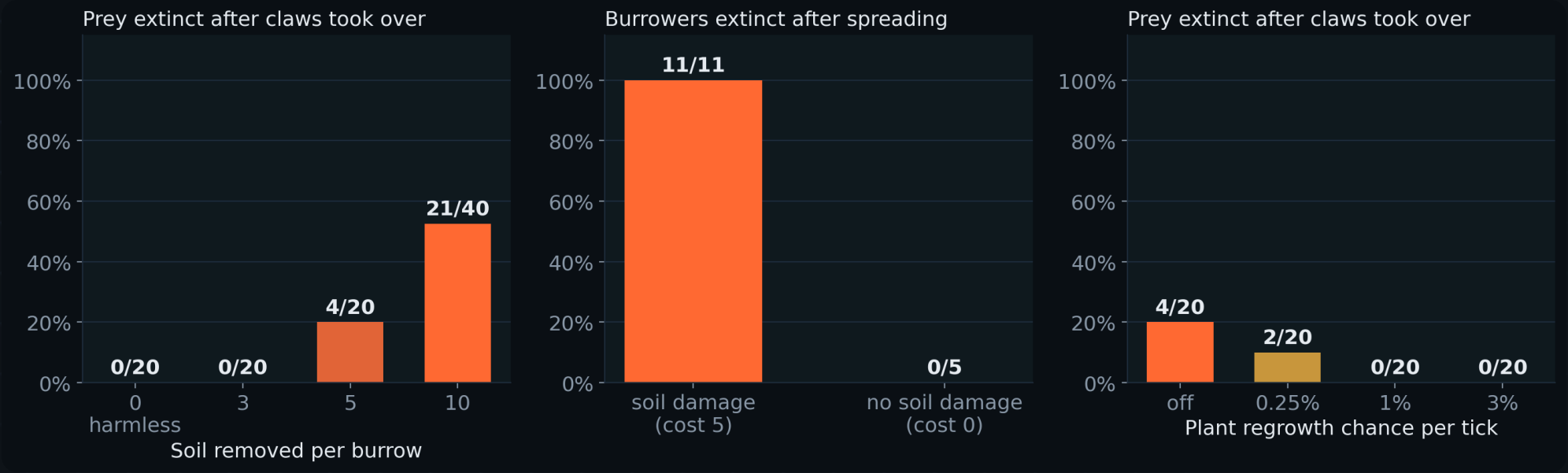


Josh's soil-layouts batch: 270 runs, 6 setups × 9 soil layouts × 5 soil seeds. Runs lasting 250,000 ticks.

Same starting species, different soil: survival ranges from 80% to 0%

QUESTION 4 · INSTRUMENTAL FEATURES AND TRAITS

What made the difference



Left: Tarique, soil removed per burrow. Middle: Nik, soil damage on vs off. Right: Jenna, plant regrowth (reference settings).

ENVIRONMENT

- **Soil damage per burrow**: 0 or 3 never collapses; 10 collapses half
- **Soil and plant regrowth**: fast recovery rescues the world
- **Food blocking**: without consumers, food chokes plant offspring

TRAITS

- **Claw**: the trait that both saves and dooms
- **Eye**: claw mutations often replace it; 11 of 12 clawed prey in one replay were blind and never burrowed
- **Killer cell**: turns clawed prey into predators; **hide length**: at 30 ticks the claw never pays

A window of settings: predators strong enough to make burrowing pay, burrows damaging enough to matter

EVOLUTION FINDS LOOPHOLES

When claws were free, predators took them, and lost their eyes



SEEDED PREDATOR

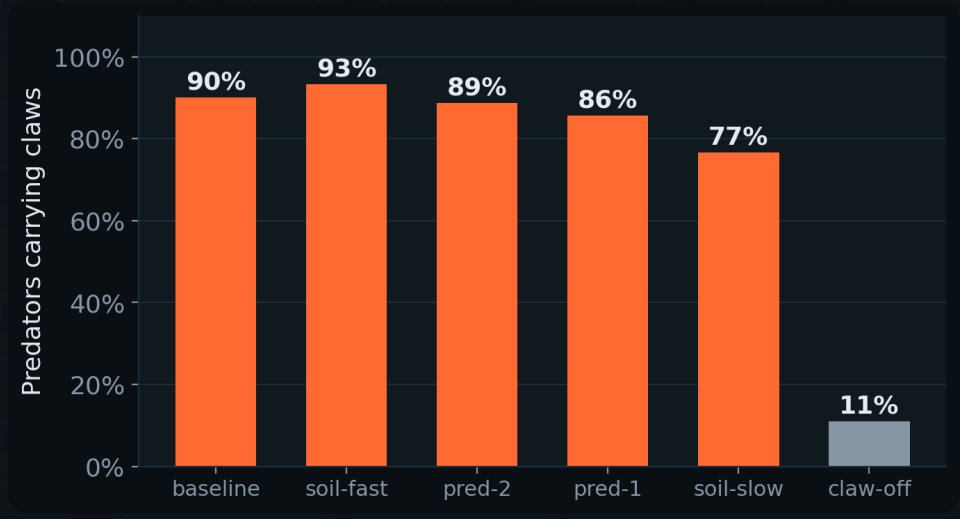
6 cells · 11 food to breed



EYELESS CLAWED PREDATOR

5 cells · 4 food to breed

- A **free claw** adds lifespan and health but costs no food, so predators keep them; none ever burrowed.
- Mouths are useless to predators and eyes cost food: a **cheap random walker** breeding ~3× as often beats a hunter.
- It came to dominate some worlds on its own: selection optimises the rules, not the design.



Share of predators carrying claws late in the runs, across 9 soil layouts. Claw-off: claws cost food.

Eyeless predator layout is illustrative; its cells (mover, killer, 3 claws) are from the species record

ANSWERS TO THE BRIEF

In four lines

1 **Did an extinction event occur?**

Yes: whole worlds, prey populations and lineages went extinct in a large share of runs.

2 **Mis-adaptation or failure to adapt?**

Mostly failure to adapt, as the first predator wave struck before the claw spread. When the claw had time, it was a mis-adaptation: favoured for hiding, fatal through the soil it destroyed.

3 **Environment vs the initial seeded species?**

The seeded predators decide whether burrowers get a chance; soil and plant recovery decide whether the burrowers then destroy themselves.

4 **Instrumental features and traits?**

Soil damage per burrow, soil and plant regrowth, food blocking; the claw, the eye it replaces, and the killer cell that turns prey into predators.

LIMITATIONS AND NEXT STEPS

What we can't claim yet

Limitations

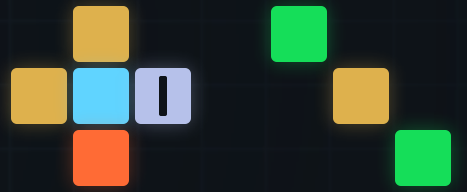
- 10–40 runs per setup: differences of 2–3 runs could be chance
- Studies used slightly different settings and grids
- Collapse rate depends on behaviour settings: with "burrow only when a predator is close" on, evolved claws collapsed 2 of 20 worlds here, against 21 of 40 in Tarique's runs with it off
- Proximate cause of death is mostly predation; the soil acts by stopping plants recovering

Next

- Measure fitness of clawed vs clawless prey **during** a decline: does selection still favour the claw?
- Replay suspected suicides with and without the invading predator
- Close the free-claw loophole for predators
- More soil maps per setup

THANK YOU

QUESTIONS?



Built on The Life Engine by Max Robinson, extended with soil fertility, burrowing claws, role tagging, seeded run setup and a headless experiment runner.

References: Gyllenberg, M. & Parvinen, K. (2001). Necessary and sufficient conditions for evolutionary suicide. *Bulletin of Mathematical Biology* 63, 981–993. · Rankin, D. J. & López-Sepulcre, A. (2005). Can adaptation lead to extinction? *Oikos* 111, 616–619. · Hardin, G. (1968). The tragedy of the commons. *Science* 162, 1243–1248.

Reports: Burrowing Into Extinction (Tarique) · Soil study (Nik) · Plant regrowth (Jenna) · Claw evolution and soil layouts (Josh)